LECTURE ACTIVITY 16: PLATE UP! – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

Learning Objectives:

By the end of this activity, you should be able to

1) State the null and the alternative hypotheses in terms of comparing population means for two categories of the explanatory variable (i.e., μ_1 and μ_2).
2) Implement the 3S strategy: find a statistic, simulate, and compute the strength of evidence against observed study results happening by chance alone.
3) When comparing two quantitative distributions, identify which has the larger mean, median, and standard deviation.
4) Conduct a simulation-based analysis when evaluating claims about two population means.
5) Find, interpret, and/or evaluate the standardized statistic and the p-value for a test of two population means
6) Conduct a theory-based analysis when evaluating claims about two population means.
7) Identify whether the validity conditions for a two-sample t-test are met.
8) Use the theory-based method to find and interpret a 95% confidence interval for the difference in population means.
9) State a complete conclusion about the alternative hypothesis (and null hypothesis) based on the magnitude of the standardized statistic and/or *p*-value and the study design.

Introduction

It's often joked that higher-end restaurants serve a minuscule amount of food on huge plates. But, is there something deeper going on? In particular, we'll address the question: **does the size of the plate on which food is served affect how much of the food gets eaten?**

The data we'll use comes from a 2019 study[1] conducted in the UK in which 134 adult volunteers, aged 18 - 61 years, were recruited and randomly allocated into one of two groups of 67 participants each. The authors note: " the sample purposefully comprised similar proportions of men and women and similar proportions of participants with lower and higher education level (no Bachelor's degree versus Bachelor's degree or higher)" (Kosīte *et al*, 2019). All participants were served the same meal (a vegetarian cheese and tomato pasta bake, though an equivalent alternate was provided, if requested); participants in one group were served the food on a small plate, whereas participants in the other group were served the food on a large plate. In addition to the plate size, the experimenters noted how many calories (in kCal) each subjects consumed (by weighing the uneaten food and using that to calculate the calories consumed).

[1] Kosīte, D., König, L.M., De-loyde, K. et al. Plate size and food consumption: a pre-registered experimental study in a general population sample. Int J *Behav Nutr Phys Act* 16, 75 (2019). https://doi.org/10.1186/s12966-019-0826-1

Question 1

(a) What is one observational unit in the context of this study?

Solution: One observational unit is one person (subject).

(b) Identify the two variables in this study. Then, classify each of them as either categorical or quantitative, and identify which variable is the explanatory variable and which is the response variable.

- **Variable 1:** **Plate Size (small or big)**
  - Classification: √ **Categorical** ◯ Quantitative
  - Type: √ **Explanatory** ◯ Response

- **Variable 2:** **Calories Consumed (in kcal)**
  - Classification: ◯ Categorical √ **Quantitative**
  - Type: ◯ Explanatory √ **Response**

(c) Is this study an observational study, random experiment, or experiment? Why?

◯ Observational Study
√ **Randomized Experiment**
◯ Experiment

Justification:

Solution: The experimenters randomly assigned subjects into groups; hence, this is a randomized experiment.

(d) Did this study make use of random assignment, random sampling, both, or neither? Explain.

Solution: As noted above, the study made use of random assignment. However, as noted in the introduction, the experimenters explicitly sought out individuals with certain attributes/characteristics, indicating that the sampling method was not (fully) random.

Addendum: CONSORT Diagrams

It's often the case in voluntary studies (like this one) that one sometimes ends up with **missing data**. For example, the authors notes "primary outcome data for one of the participants were missing due to an administrative error" (Kosīte *et al*, 2019) meaning even though the study started out with two groups of 67 subjects each, final data was only recorded on 133 subjects.

To help keep track of things like this, experimental designers often make use of **CONSORT Diagrams**. CONSORT (more accurately stylized as ConSORT) is actually an acronym: **Con**solidated **S**tandards **o**f **R**eporting **T**rials. Shown in Figure (1) is the exact CONSORT diagram that appears in the original article.

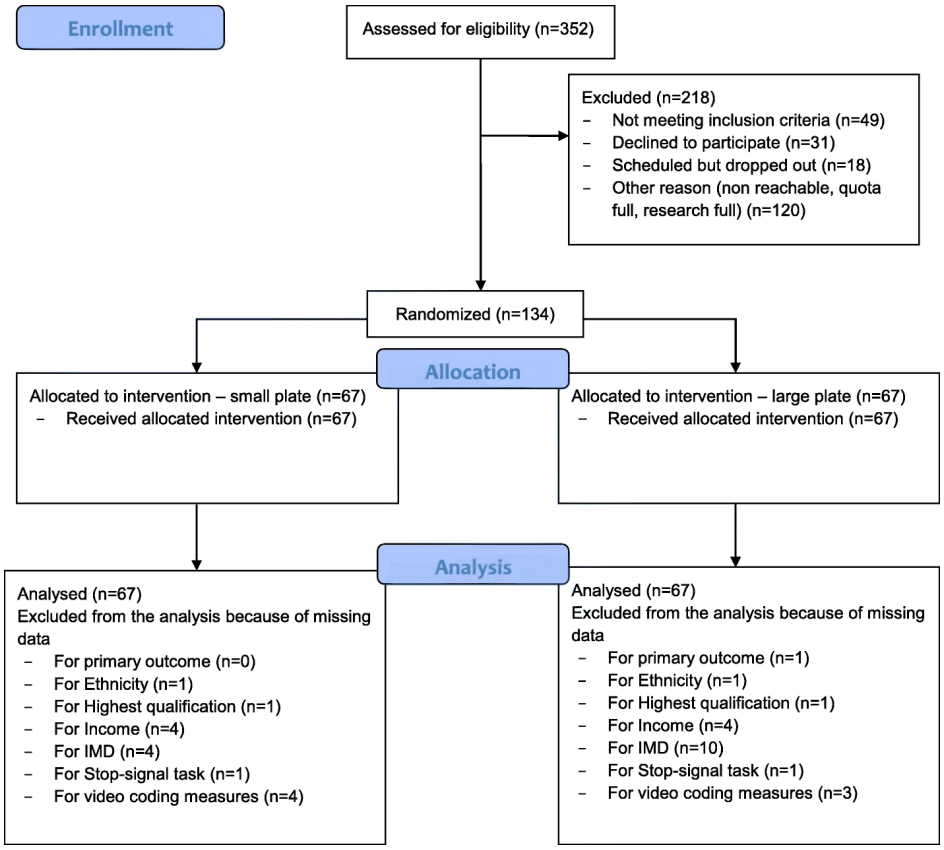


Figure 1: CONSORT Diagram from the original article. All rights retained by the original authors.

For example, the CONSORT diagram shows a few interesting things:

- The experimenters actually assessed 352 people for eligibility, before landing on the final 134
- The one person whose response values were lost due to administrative errors was a member of the "large plate" group

I almost guarantee you'll encounter at least one CONSORT diagram in your future careers!

Okay, back to the study.

Question 2

(a) State the null and the alternative hypotheses in symbols, to investigate whether plate size affects the amount of calories consumed (among UK residents similar to those included in the study).

Solution:

- **Null Hypothesis:** $H_0 : \mu_{small} = \mu_{big}$ or, equivalently, $H_0 : \mu_{small} - \mu_{big} = 0$
- **Alternative Hypothesis:** $H_A : \mu_{small} \neq \mu_{big}$ or, equivalently, $H_A : \mu_{small} - \mu_{big} \neq 0$

(b) Define the symbols you used in the hypotheses above.

Solution: The symbols represent the following population parameters:

- μ_{small}: the population mean amount of calories (in kCal) that would be consumed by UK residents (similar to those included in the study) who are served the pasta bake on the small plate
- μ_{big}: the population mean amount of calories (in kCal) that would be consumed by UK residents (similar to those included in the study) who are served the pasta bake on the small plate

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Step 3: Explore the Data

You may notice - this lecture activity is not one that appears in the textbook. That is to say, the data won't come pre-loaded in any of the applets we'll be using today. That's okay! Indeed, part of the reason I wanted to use this particular dataset is to give you practice with **uploading** data to the applets (as this is something you'll need to do for your final projects...)

Carry out the following steps:

1) Navigate to the Data File

2) Select all (Control/Command + A), and copy (Control/Command + C)

3) Navigate to the Multiple Means applet (this is the same "Two Means" applet we used in Lecture Activity 15, "Lingering Effects of Sleep Deprivation")

4) Press **Clear**, click inside the sample data box, and paste the data (Control/Command + V)

5) Click **Use Data**

Question 3

(a) Report the sample size, sample mean, and sample SD of the amount of calories (in kCal) consumed within each group.

Small Plate Group	Big Plate Group
$n_{small} =$ 67	$n_{big} =$ 66
$\bar{x}_{small} =$ 624.91 kCal	$\bar{x}_{big} =$ 644.12 ms
$s_{small} =$ 292.34 ms	$s_{big} =$ 264.96 ms

Table 1: Summary Statistics

(b) Calculate the statistic (observed difference in the sample mean number of close friends between these two groups), and confirm this matches with the output from the applet (in the "Observed Difference in means" field).

$$\bar{x}_{\text{small}} - \bar{x}_{\text{big}} = \underline{\textbf{-19.203 kCal}}$$

(c) Note that is is impossible to consume a negative amount of calories. So, why is the difference in Part (b) negative?

Solution: The negative value in Part (b) is a *difference* in average calories consumed. Since the small plate sample mean is big plate sample mean, the subtraction produces a negative value. The statistic is a difference between sample means, so it can be negative even though individual calorie consumption cannot be negative.

Step 4: Draw Inferences Beyond the Data

The question we want to answer is, *Is the difference in mean amount of calories consumed between the small-plate and big-plate groups as seen among the 133 subjects something that could plausibly have happened by chance, by random sampling alone, if, on average, plate size has no effect on calorie consumption?*

Simulation-Based Approach (Review)

Let's quickly run through a simulation-based approach, using the same steps we did in Lecture Activity 15.

1. In the applet, check the **Show Shuffle Options** box
2. Enter 10,000 for the **Number of Shuffles**, and click the "**Plot**" option under "**Select display**"
3. Press the **Shuffle Responses** button

Question 4

(a) Use the applet to find and report the simulation-based p-value. To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) *p*-value.

- My approximate simulation-based *p*-value is **0.6947** **(answers may vary slightly)**

Here is my screenshot:

Solution:

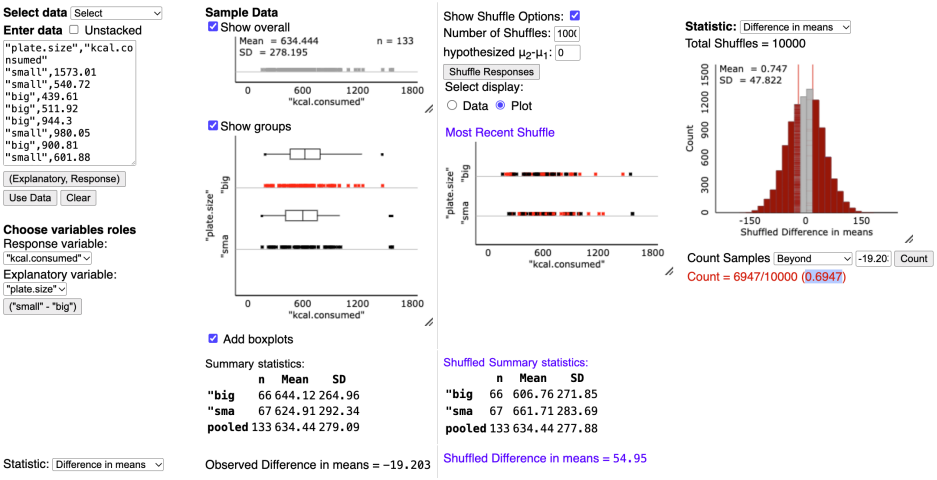


Figure 2: Screenshot from the Multiple Means Applet

(b) ***Interpret*** this *p*-value.

Solution: The probability of obtaining a difference in sample means of -19.203 kCal or more extreme values in either direction merely from random sampling is about 0.6947, assuming that the average amount of calories consumed within the small-plate group is the same as the average amount of calories consumed within the big-plate group across all UK adults (similar to those included in the study)

(c) ***Evaluate*** this *p*-value: How much evidence do the sample data provide against the null hypothesis? Choose one.

- √ **Not much evidence against the null hypothesis**
- ◯ Moderate evidence against the null hypothesis
- ◯ Strong evidence against the null hypothesis
- ◯ Very strong evidence against the null hypothesis

Theory-Based Approach (Two-Sample *t*-Test)

Another measure of the strength of evidence against the null hypothesis would be to standardize the statistic by subtracting the hypothesized value and dividing by the standard error of the statistic.

Navigate back to the Multiple Means applet, and use the "**Statistic**" dropdown menu (below the sample data box) to select "**t-statistic**."

Question 5

Record the value of the (simulation-based) *t*-statistic, and interpret this value.

Solution: The observed *t*-statistic is $t = \mathbf{-0.40}$. The difference in sample means is about 0.40 standard deviations below the null value of 0.

Notice that the standardized statistic uses the letter *t* (the one you should remember from Chapter 2) instead of the *z* (from Chapters 1 and 5) we saw when testing proportions. This is because the theoretical distribution used is now a *t-distribution* instead of a normal distribution. These t-distributions are very similar to normal distributions, especially when sample sizes are large. The *t*-statistic, like the *z*-statistic, tells us how many standard deviations our difference in sample means is above or below the hypothesized mean of the null distribution and it can be judged in the same manner.

Question 6

In light of the value of the standardized statistic, should you expect the *p*-value to be large or small? How are you deciding?

Solution: The *p*-value should be large because a *t*-statistic of -0.40 lies within two standard deviations of the mean of 0. Under the null hypothesis, values this far from 0 in either direction are fairly likely to happen.

Question 7

The histogram on the far right now displays the null distribution of simulated *t*-statistics. Describe this distribution.

Solution: The distribution is unimodal and bell-shaped, centered near zero (technically, centered at 0.016), with a standard deviation of 0.998

Question 8

Below the null distribution check the box "**Overlay t distribution**". Does the *t*-distribution appear to adequately predict the behavior of the shuffled *t*-statistics? What do you think this suggests about whether the validity conditions will be met for these data?

Solution: Yes. The theoretical *t*-distribution closely follows the null distribution of shuffled *t*-statistics. This agreement suggests that the theory-based model is appropriate and that the validity conditions are satisfied for these data.

Question 9

(a) Calculate the theory-based *p*-value by entering the observed value of the t-statistic in the Count Samples box and press Count. (Remember to use the pull-down menu to specify what direction(s) you want to consider more extreme based on the alternative hypothesis.) **To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) statistic and (3) *p*-value.**

- My approximate simulation-based *p*-value is **0.6898**

Here is my screenshot:

Solution:

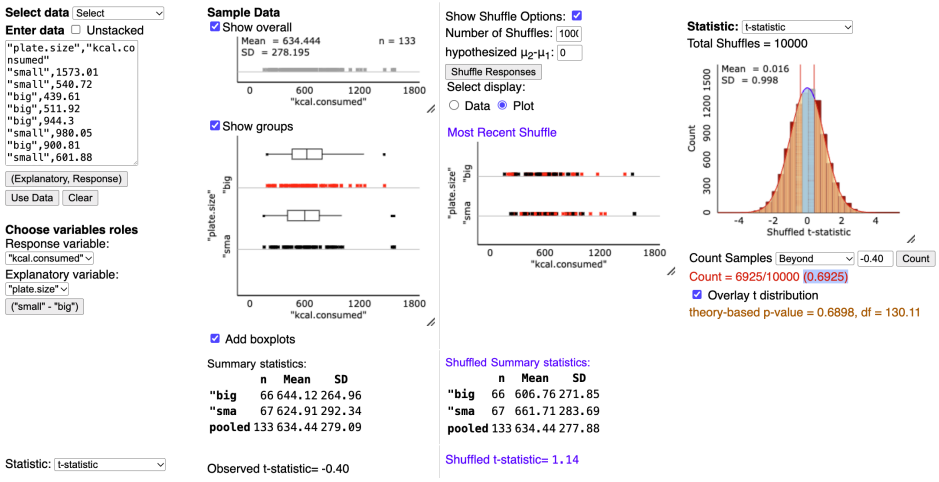


Figure 3: Screenshot from the Multiple Means Applet

(b) How does the theory-based *p*-value from the *t*-distribution (in orange) compare to the simulation-based *p*-value from above?

Solution: The theory-based *p*-value of 0.6898 is very close to the simulation-based *p*-value of 0.6925

Concept Overview

The **validity conditions** required for this theory-based approach (a "**two-sample *t*-test**") to be valid that the quantitative variable should have a symmetric distribution in both groups or you should have at least 20 observations in each group and the sample distributions should not be strongly skewed.

Question 10

Do the validity conditions appear to be satisfied for these data? Justify your answer, possibly referring to additional outputs from the applet if necessary.

Solution: The parallel boxplots from the applet reveal that the distribution of response values in the "big" group may be slightly right-skewed, and that both groups have a couple of outliers. As such, we may have some small concerns about the validity conditions - however, the sample sizes in each group are relatively large (66 and 67), so we should be okay. (Indeed, we already saw that the theory-based and simulation-based *p*-values are very close to one another!)

As we've seen before, the Theory-Based Inference applet can also be used to conduct theory-based inference. Shown below, in Figure (4), is a screenshot from the Theory-Based Inference applet demonstrating the theory-based approach:

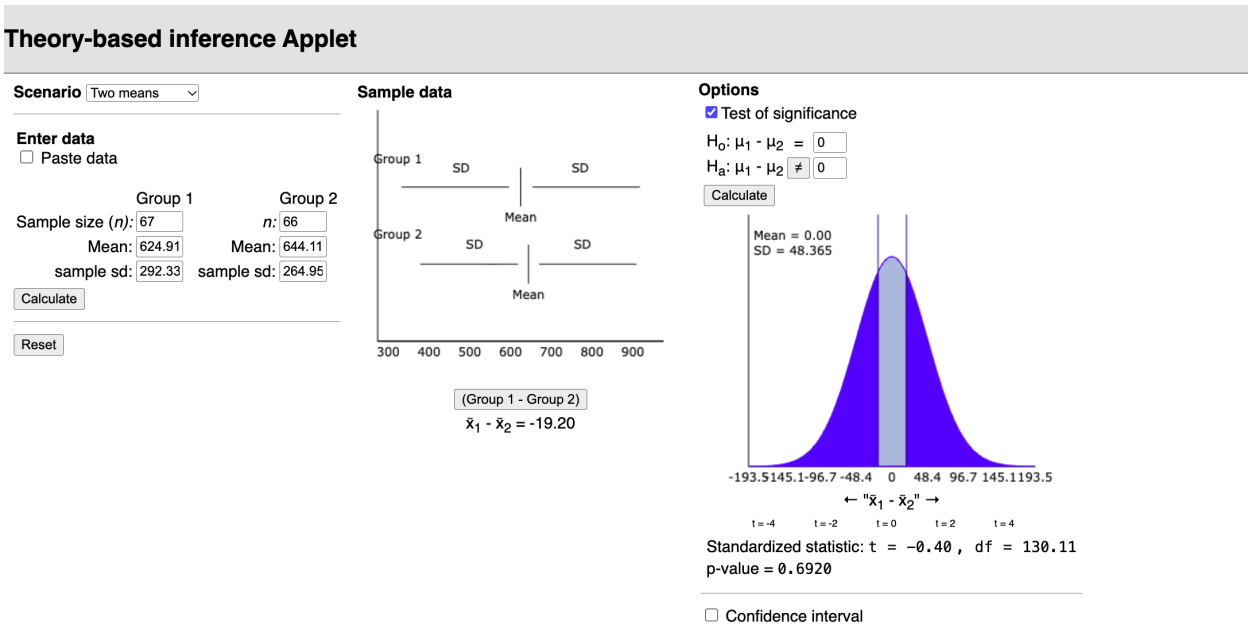


Figure 4: Screenshot from the Theory-Based Inference Applet

Indeed, we obtain the same *t*-statistic as from the Multiple Means applet and nearly the same *p*-value.

The applet automatically gave us the *t*-statistic, but as we learned previously we could calculate the test statistic with the following formula:

$$t = \frac{(\text{statistic}) - (\text{hypothesized mean of null distribution})}{\text{SD}_{\text{NULL}}} = \frac{\text{-19.20} - 0}{48.365} \approx \textbf{-0.397}$$

Step 5: Formulate Conclusions

Based on the analysis you have just carried out, we can state out conclusion in the context of the study. Remember, we need to comment on:

Question 11

***Statistical significance*:** Based on your analysis, give an overall conclusion in context. Make sure you refer to the parameter of interest in your conclusion.

Solution: There is not much evidence to suggest that the population average amount of calories consumed by those presented with food on a small plate is different from the population average amount of calories consumed by those presented with food on a big plate, among UK residents similar to those included in the study.

Question 12

***Generalization*:** To whom can you apply the results of this study? All people? All adults? How are you deciding?

Solution: Even though the sample is not random, the introduction reveals that the authors took care to obtain a *representative* sample. As such, we can probably generalize to the set of all UK residents aged 18 - 61 years.

Question 13

***Causation*:** Can we make a causal claim here? If so, what is it? If not, why not?

Solution: Even though this is a randomized experiment, we did not find sufficient evidence to suggest an association between plate size and calories consumed exists. As such, even though we'd be justified in making a causal claim, there isn't really a one to make here.

Estimation

We will now use the Theory-based Inference Applet to calculate the 95% confidence interval.

1) Go to the Theory-Based Inference applet

2) Use the pull-down menu to select **Two Means**

3) Enter sample size, sample mean, and sample standard deviation for each group.

4) Click **Calculate**.

5) Check the box next to **Confidence Interval**, and click **Calculate CI**

Question 14

(a) Report the 95% confidence interval for this parameter.

Solution: (-114.8894, 76.4792)

(b) Does the 95% confidence interval calculated from the sample data contain the value 0? What does that imply? (**Hint**: Recall that a confidence interval is an interval of plausible values for the parameter of interest, and the interval is calculated using the sample statistics.)

Solution: Zero is contained in the interval, so it is a plausible value for the difference in population means.This supports the conclusion that the population means may not actually differ.

(c) Does the 95% confidence interval agree with your conclusion above? How are you deciding?

Solution: Yes. The interval agrees with the conclusion above because it contains zero. This indicates that the population mean calories consumed by those presented the food on a small plate could be the same as the population mean calories consumed by those presented the food on a big plate.

(d) Fill in the blanks of the sentence to complete the interpretation of the 95% confidence interval:

We are 95% confident that the difference in **population mean** (population mean/sample mean) amount of calories consumed by all people presented with the food on a small plate and the **population mean** (population mean/sample mean) amount of calories consumed by all people presented with the food on a big plate is between **-114.8894 kCal** and **76.4792 kCal**.

Step 6: Look Back and Ahead

Question 15

(a) Did anything about the design and conclusions of this study concern you? Issues you may want to critique include:

- The match between the research question and the study design
- How the observational units were selected
- How the measurements were recorded
- The number of observational units in the study
- Whether what we observed is of practical value

Solution: Answers may vary! Here are my thoughts: overall, I think this study was conducted quite well; the research question is well-matched by the study, and the sample sizes were fairly large. Even with regards to the sampling method, I think the authors did a good job; simple random sampling is often fairly difficult to implement in practice, so the next best thing to do is to try and obtain as representative a sample as possible - something the researchers took care to do.

(b) What should the researchers' next steps be to fix the limitations in this study and/or build on this knowledge?

Solution: It may be interesting to see the results of this study conducted on different foods, and to see if there is any variability in calories consumed across different foods.